

- Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33):E7665–E7671, 2018. ISSN 0027-8424. doi: 10.1073/pnas.1806579115. URL <https://www.pnas.org/content/115/33/E7665>.
- Alireza Mousavi-Hosseini, Sejun Park, Manuela Girotti, Ioannis Mitliagkas, and Murat A Erdogdu. Neural networks efficiently learn low-dimensional representations with sgd. *arXiv preprint arXiv:2209.14863*, 2022.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability, 2023.
- Gadi Naveh and Zohar Ringel. A self consistent theory of gaussian processes captures feature learning effects in finite cnns. *Advances in Neural Information Processing Systems*, 34, 2021.
- Gadi Naveh, Oded Ben David, Haim Sompolinsky, and Zohar Ringel. Predicting the outputs of finite deep neural networks trained with noisy gradients. *Physical Review E*, 104(6), Dec 2021. ISSN 2470-0053. doi: 10.1103/physreve.104.064301. URL <http://dx.doi.org/10.1103/PhysRevE.104.064301>.
- Radford M Neal. Priors for infinite networks. In *Bayesian Learning for Neural Networks*, pp. 29–53. Springer, 1996.
- Radford M Neal et al. MCMC using hamiltonian dynamics. *Handbook of markov chain monte carlo*, 2(11):2, 2011.
- Roman Novak, Lechao Xiao, Jaehoon Lee, Yasaman Bahri, Greg Yang, Daniel A. Abolafia, Jeffrey Pennington, and Jascha Sohl-Dickstein. Bayesian Deep Convolutional Networks with Many Channels are Gaussian Processes. *arXiv e-prints*, art. arXiv:1810.05148, October 2018.
- Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.
- Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning (Adaptive Computation and Machine Learning)*. The MIT Press, 2005. ISBN 026218253X.
- David Saad and Sara A. Solla. Exact solution for on-line learning in multilayer neural networks. *Phys. Rev. Lett.*, 74:4337–4340, May 1995. doi: 10.1103/PhysRevLett.74.4337. URL <https://link.aps.org/doi/10.1103/PhysRevLett.74.4337>.
- Stefano Sarao Mannelli, Eric Vanden-Eijnden, and Lenka Zdeborová. Optimization and generalization of shallow neural networks with quadratic activation functions. *Advances in Neural Information Processing Systems*, 33:13445–13455, 2020.
- Inbar Seroussi, Gadi Naveh, and Zohar Ringel. Separation of scales and a thermodynamic description of feature learning in some CNNs. *Nature Communications*, 14(1):908, 2023.
- H. S. Seung, H. Sompolinsky, and N. Tishby. Statistical mechanics of learning from examples. *Phys. Rev. A*, 45:6056–6091, Apr 1992. doi: 10.1103/PhysRevA.45.6056. URL <https://link.aps.org/doi/10.1103/PhysRevA.45.6056>.
- Yan Shuo Tan and Roman Vershynin. Online stochastic gradient descent with arbitrary initialization solves non-smooth, non-convex phase retrieval. *arXiv preprint arXiv:1910.12837*, 2019.
- Naftali Tishby and Noga Zaslavsky. Deep learning and the information bottleneck principle, 2015.
- David Tong. *Statistical physics*. University of Cambridge, 2011.
- Vikrant Varma, Rohin Shah, Zachary Kenton, János Kramár, and Ramana Kumar. Explaining grokking through circuit efficiency. *arXiv preprint arXiv:2309.02390*, 2023.
- Bojan Žunkovič and Enej Ilievski. Grokking phase transitions in learning local rules with gradient descent. *arXiv e-prints*, art. arXiv:2210.15435, October 2022. doi: 10.48550/arXiv.2210.15435.

Karl Weiss, Taghi M Khoshgoftaar, and DingDing Wang. A survey of transfer learning. *Journal of Big data*, 3(1):1–40, 2016.

Christopher Williams. Computing with infinite networks. *Advances in neural information processing systems*, 9, 1996.

Zhao You, Xiaorui Wang, and Bo Xu. Exploring one pass learning for deep neural network training with averaged stochastic gradient descent. In *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 6854–6858, 2014. doi: 10.1109/ICASSP.2014.6854928.

Matthew D. Zeiler and Rob Fergus. Visualizing and understanding convolutional networks. In David Fleet, Tomas Pajdla, Bernt Schiele, and Tinne Tuytelaars (eds.), *Computer Vision – ECCV 2014*, pp. 818–833, Cham, 2014. Springer International Publishing. ISBN 978-3-319-10590-1.

# SUPPLEMENTARY MATERIAL

## A TEACHER-STUDENT MODEL

### A.1 SCALING LIMIT OF TEACHER-STUDENT MODEL AND SAMPLE COMPLEXITY ASPECTS

Here we detail the scaling limits in which we believe our results become exact. To this end, we first point to the factors controlling the three approximations we employ and discuss scaling limits in which these factors all vanish.

Our first approximation is the mean-field decoupling between the read-out layer and the input layer. Specifically we follow Mei et al. (2018); Seroussi et al. (2023); Bordelon & Pehlevan (2022), and introduce a mean-field scaling parameter  $\chi$  define  $\sigma_a^2 = \tilde{\sigma}_a^2/\chi$  (i.e. the microscopic variance of each readout layer weight is  $\tilde{\sigma}_a^2/(\chi N)$ ) and  $\sigma^2 = \tilde{\sigma}^2/\chi$  (i.e. the noise variance on the gradients is  $\sigma^2$ ). Notably, since  $b, c \propto \sigma^2/[\sigma^2 + K]$  and  $K \propto \sigma_a^2$ , we find that  $\chi$  has no effect on  $b$  or  $c$ . Examining the action, this yields an additional constant factor of  $\chi$  in front of the non-linear term, as expected since mean-field scaling increases feature learning. Taking  $\chi \gg 1$  controls the accuracy of our mean-field decoupling between the read-out and input layer.

Our second approximation is the continuum limit, where we trade summation over data-points with integrals. This approximation is closely related to the EK approximation Rasmussen & Williams (2005); Cohen et al. (2021). Taking  $n$  to be large while keeping  $n/(\tilde{\sigma}^2) \equiv n_{\text{eff}}$  fixed, corrections to this approximation scale as  $1/\tilde{\sigma}^2$ . Notably taking large  $\tilde{\sigma}^2$  while keeping  $n_{\text{eff}}$  fix does not change  $b, c$  or the action of the weights as the latter contains only the combination  $n/\tilde{\sigma}^2$ .

The third approximation we carry is the saddle-point approximation or Gaussian-mixture approximation for  $p(\mathbf{w})$ . Examining our action

$$\mathcal{S}[\mathbf{w}] = d \left( \frac{|\mathbf{w}|^2}{2\sigma_w^2} - \frac{2\chi n_{\text{eff}} \tilde{\sigma}_a^2}{\pi d N} \frac{(\mathbf{w} \cdot \mathbf{w}^*)^2}{1 + 2|\mathbf{w}|^2} \left( b[n_{\text{eff}}/d] - \frac{2c[\epsilon] (\mathbf{w} \cdot \mathbf{w}^*)^2}{1 + 2|\mathbf{w}|^2} \right)^2 \right) \quad (1)$$

where we further indicated the dependencies of  $b$  and  $c$  which arise through GPR,  $d$  comes out as controlling the saddle point approximation, provided the other coefficients in the action do not scale down with  $d$ .

We thus find that the following family of scaling limits controlled by two scaling parameters ( $\alpha, \beta \rightarrow \infty$ ) ensures  $\mathcal{S}[\mathbf{w}]$  is exact and its saddle points dominate  $\mathbf{w}$

$$\begin{aligned} N &\rightarrow \beta N \\ d &\rightarrow \sqrt{\beta} d \\ \chi &\rightarrow \sqrt{\beta} \chi \\ \tilde{\sigma}^2 &\rightarrow \frac{\alpha}{\sqrt{\beta}} \tilde{\sigma}^2 \quad \sigma^2 \rightarrow \frac{\alpha}{\beta} \sigma^2 \\ n &\rightarrow n\alpha \end{aligned} \quad (2)$$

Notably increasing  $\alpha$  makes the continuum approximation exact Cohen et al. (2021) whereas increasing  $\beta$  makes both the saddle point and the mean-field decoupling exact. Furthermore, it can be verified that both  $\alpha$  and  $\beta$  have no effect on  $\tilde{S}$  and in particular keep the coefficient in front of the non-linear term invariant. The only conflict between these two limits is in  $\tilde{\sigma}^2$  hence we require  $\alpha^2 > \beta$ , but otherwise one is free to scale these up differently.

We conjecture that the teacher-student model has a better sample complexity than the infinite-width network described by the GP limit. Namely, the GP limit is able to learn the cubic target at,  $n \propto d^3$  whereas the former can reach good accuracy at  $n \propto d$ . Replacing  $n$  with the above,  $n_{\text{eff}}$  the latter is implied by the linear scaling of  $d, n_{\text{eff}}$  together with our analytical results, which show that the cubic component is being learned. In general, reduction of complexity has been seen in the context of Gradient descent in Bietti et al. (2022); Sarao Mannelli et al. (2020); Gamarnik et al. (2019) as well as in online learning up to logarithmic factors Arous et al. (2021); Mousavi-Hosseini et al. (2022); Tan & Vershynin (2019).

Establishing this more rigorously requires showing that  $n$  itself, and not  $n/\bar{\sigma}^2$  can be kept proportional to  $d$ . Alternatively stated, that  $\alpha$  can scale as  $\sqrt{\beta}$  while introducing corrections to the theory proportional to  $\sqrt{\beta}/\alpha$ , which one can then take to be as small as desired. This scenario seems to be consistent with the corrections to EK derived in Ref. Cohen et al. (2021), which when evaluated in simple settings (random data on a hypersphere and a rotationally invariant kernel) indeed show that  $\sqrt{\beta}/\alpha$  controls corrections to EK.

## A.2 CONTINUUM ACTION AND MODE DECOUPLING

Our first step here consists of removing the randomness induced by the specific choice of training set. To this end, we consider taking a large  $n$  limit while keeping  $n/\sigma^2$  fixed. This has several merits. At large,  $n$  it is natural to replace summations in Eq. (7) by integrals, which can be evaluated directly. Furthermore, in this limit, one obtains an analytical handle on Gaussian processes regression via the equivalence kernel (EK) approximation Rasmussen & Williams (2005); Cohen et al. (2021). Notably, previous works have found reasonable convergence to this approximation, in similar scenarios to ours, for values as low as  $n = 100$  and  $\sigma^2 = 0.1$  Cohen et al. (2021). A key element in the EK approximation is the spectrum of the kernel defined by

$$\int d\mu_z Q(\mathbf{x}, \mathbf{z}) \phi_\lambda(\mathbf{z}) = \lambda \phi_\lambda(\mathbf{x}) \quad (3)$$

where  $\mu_z$  is the dataset measure, which corresponds here to i.i.d. standard Gaussians. Denoting by  $g_\lambda$  the spectral decomposition of the target function on  $\phi_\lambda(\mathbf{x})$ , the GPR predictions (Eq. 9) are given by

$$\bar{f}(\mathbf{x}) = \sum_\lambda \frac{\lambda}{\lambda + \sigma^2/n} g_\lambda \phi_\lambda(\mathbf{x}) \quad (4)$$

where  $\frac{\lambda}{\lambda + \sigma^2/n}$  are the learnability factors. Notably,  $n$  only enters through the combination,  $n/\sigma^2$  which we treat as the effective amount of data. Hence, we can control the amount of learning in two equivalent ways, by either providing more data points or by reducing the noise.

The resulting limit also enables us to identify the two relevant directions in function space, being  $H_1(\mathbf{x})$  and  $H_3(\mathbf{x})$ . Specifically we argue that at large  $d$  the operator  $Q(\mathbf{x}, \mathbf{y})$  is block-diagonal within the space spanned by  $H_1(\mathbf{x})$  and  $H_3(\mathbf{x})$ . Since it acts on a target having support only on  $H_1(\mathbf{x}), H_3(\mathbf{x})$ , this limits  $\bar{f}(\mathbf{x})$  to this space and hence also  $\bar{t}(\mathbf{x}) = [y(\mathbf{x}) - \bar{f}(\mathbf{x})]/\sigma^2$ . Our claim is shown to be self-consistent in App. A namely, that making this assumption results in  $Q$  and  $\bar{t}$  which obey the same assumption. Operationally, we thus take the ansatz

$$\sigma^2 \bar{t}(\mathbf{x}) = b H_1(\mathbf{x}) + c H_3(\mathbf{x}) \quad (5)$$

where  $b, c \in \mathbb{R}$  are some unknown constant coefficients. Later, we derive self-consistent equations determining these constants. Given the above ansatz, we turn to the action in Eq. 7, take the continuum approximation where  $\sum_{\mu=1}^n \rightarrow n \int d\mu_y$  and find that the integral appearing in the action can be solved exactly-  $\bar{t}(\mathbf{x}) = b H_1(\mathbf{w}^* \cdot \mathbf{x}) + c H_3(\mathbf{w}^* \cdot \mathbf{x})$ , we can compute the action in the continuum limit-

$$\begin{aligned} \mathcal{S} &\equiv \frac{d|\mathbf{w}|^2}{2\sigma_w^2} - \frac{n^2\sigma_a^2}{2N\sigma^4} \left( \underbrace{\left( b - 3c|\mathbf{w}^*|^2 \right) \int d\mu_x (\mathbf{w}^* \cdot \mathbf{x}) \phi(\mathbf{w} \cdot \mathbf{x})}_{I_0} + c \underbrace{\int d\mu_x (\mathbf{w}^* \cdot \mathbf{x})^3 \phi(\mathbf{w} \cdot \mathbf{x})}_{I_1} \right)^2 \\ &= d \left( \frac{|\mathbf{w}|^2}{2\sigma_w^2} - \frac{2n^2\sigma_a^2}{\pi d N \sigma^4} \frac{(\mathbf{w} \cdot \mathbf{w}^*)^2}{1 + 2|\mathbf{w}|^2} \left( b - \frac{2c(\mathbf{w} \cdot \mathbf{w}^*)^2}{1 + 2|\mathbf{w}|^2} \right)^2 \right) \end{aligned} \quad (6)$$

we the integrals  $I_0, I_1$  are computed in App. A.4.

In principle, one could perform saddle point analysis in two variables. A simplification can be made by noting that the dominant fluctuations are in the direction of  $\mathbf{w}^*$ . We are therefore projecting

$\mathbf{w}$  into the subspace of  $\mathbf{w}^*$ , such that,  $\mathbf{w} = P_{\mathbf{w}^*} \mathbf{w} + P_{\mathbf{w}^*}^\perp \mathbf{w}$ , with  $P_{\mathbf{w}^*} = \frac{1}{|\mathbf{w}^*|^2} \mathbf{w}^* (\mathbf{w}^*)^T$ , and  $P_{\mathbf{w}^*}^\perp = I_d - P_{\mathbf{w}^*}$ . Denote  $\alpha = \mathbf{w}^* \cdot \mathbf{w}$  and  $\alpha_\perp = |P_{\mathbf{w}^*}^\perp \mathbf{w}|$  the action is given by

$$\mathcal{S}[\mathbf{w}] = \frac{d}{2\sigma_w^2} \alpha^2 + \frac{d}{2\sigma_w^2} \alpha_\perp^2 - \frac{4n^2 \sigma_a^2}{\pi 2N \sigma^4} \frac{|\mathbf{w}^*|^2 \alpha^2}{1 + 2(\alpha^2 + \alpha_\perp^2)} \left( b - \frac{2c |\mathbf{w}^*|^2 \alpha^2}{1 + 2(\alpha^2 + \alpha_\perp^2)} \right)^2, \quad (7)$$

By concentration of the norm, we replace  $\alpha_\perp$  by its average  $\sigma_w^2$  yields the following action-

$$\tilde{\mathcal{S}}[\mathbf{w} \cdot \mathbf{w}^*] = d \left( \frac{(\mathbf{w} \cdot \mathbf{w}^*)^2}{2\sigma_w^2} - \frac{2n^2 \sigma_a^2}{\pi \sigma^4 d N} \frac{(\mathbf{w} \cdot \mathbf{w}^*)^2}{1 + 2(\sigma_w^2 + (\mathbf{w} \cdot \mathbf{w}^*)^2)} \left( b - \frac{2c (\mathbf{w} \cdot \mathbf{w}^*)^2}{1 + 2(\sigma_w^2 + (\mathbf{w} \cdot \mathbf{w}^*)^2)} \right)^2 \right) \quad (8)$$

### A.3 SOLVING THE EQUATIONS FOR $b, c$

Since the kernel does not mix polynomials of higher order, we can write the GPR expression for the mean network output in the function space spanned by the components of the target. Namely, any function in this two-dimensional function space is given by a linear combination of  $\hat{H}_1(\mathbf{x}) = H_1(\mathbf{x})$ ,  $\hat{H}_3(\mathbf{x}) = \frac{1}{\sqrt{6}} H_3(\mathbf{x})$ . The factor of  $\frac{1}{\sqrt{6}}$  results from the requirement that the basis of this space will be normal. In this basis, the discrepancy and the target is given by,

$$\bar{\mathbf{t}} = \begin{bmatrix} b \\ \sqrt{6}c \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 1 \\ \sqrt{6}\epsilon \end{bmatrix} \quad (9)$$

In this basis, the two dimensional kernel  $\tilde{Q}$  is given by

$$\tilde{Q} = \begin{bmatrix} \int d\mu_{\mathbf{x}} d\mu_{\mathbf{z}} \hat{H}_1(\mathbf{x}) Q(\mathbf{x}, \mathbf{z}) \hat{H}_1(\mathbf{z}) & \int d\mu_{\mathbf{x}} d\mu_{\mathbf{z}} \hat{H}_3(\mathbf{x}) Q(\mathbf{x}, \mathbf{z}) \hat{H}_1(\mathbf{z}) \\ \int d\mu_{\mathbf{x}} d\mu_{\mathbf{z}} \hat{H}_1(\mathbf{x}) Q(\mathbf{x}, \mathbf{z}) \hat{H}_3(\mathbf{z}) & \int d\mu_{\mathbf{x}} d\mu_{\mathbf{z}} \hat{H}_3(\mathbf{x}) Q(\mathbf{x}, \mathbf{z}) \hat{H}_3(\mathbf{z}) \end{bmatrix} \quad (10)$$

where  $\mu_{\mathbf{x}}$  is the standard Gaussian measure,  $Q$  is the mean continuum limit kernel defined as

$$Q(\mathbf{x}, \mathbf{z}) = \sigma_a^2 \int \frac{e^{-\mathcal{S}(\mathbf{w}; b, c)}}{\mathcal{Z}} \phi(\mathbf{x} \cdot \mathbf{w}) \phi(\mathbf{z} \cdot \mathbf{w}) d\mathbf{w} \quad (11)$$

Next, we proceed by calculating the matrix element of  $\tilde{Q}$ . As shown in the integral appendix A.4, we have

$$\begin{aligned} \int d\mu_{\mathbf{x}} \hat{H}_1(\mathbf{x}) \phi(\mathbf{x} \cdot \mathbf{w}) &= \frac{2(\mathbf{w}^* \cdot \mathbf{w})}{\sqrt{\pi} \sqrt{(1 + 2|\mathbf{w}|^2)}} \\ \int d\mu_{\mathbf{x}} \hat{H}_3(\mathbf{x}) \phi(\mathbf{x} \cdot \mathbf{w}) &= -\frac{4(\mathbf{w}^* \cdot \mathbf{w})^3}{\sqrt{6\pi} \sqrt{(1 + 2|\mathbf{w}|^2)^3}} \end{aligned}$$

Since the fluctuations in the directions orthogonal to  $\mathbf{w}^*$  are small, the quantity  $|\mathbf{w}|^2$  appearing in the denominator can be approximated to  $(\mathbf{w} \cdot \mathbf{w}^*)^2 + (d-1) \frac{\sigma_w^2}{d}$  as explained in the main text. Since we are taking large values of  $d$ , then  $\frac{d-1}{d} \cong 1$ . Additionally, we denote  $q = \mathbf{w} \cdot \mathbf{w}^*$ , so that .

$$\tilde{Q} = \frac{\sigma_a^2}{\mathcal{Z}} \begin{bmatrix} \int \frac{2q^2 e^{-\mathcal{S}(q; b, c)}}{\pi(1 + 2(q^2 + \sigma_w^2))} dq & - \int \frac{8q^4 e^{-\mathcal{S}(q; b, c)}}{\sqrt{6\pi}(1 + 2(q^2 + \sigma_w^2))^2} dq \\ - \int \frac{8q^4 e^{-\mathcal{S}(q; b, c)}}{\sqrt{6\pi}(1 + 2(q^2 + \sigma_w^2))^2} dq & \int \frac{8q^6 e^{-\mathcal{S}(q; b, c)}}{3\pi(1 + 2(q^2 + \sigma_w^2))^3} dq \end{bmatrix}$$

Where  $\mathcal{Z} = \int dq e^{-\mathcal{S}(q; b, c)}$ . These integrals are computed numerically, given,  $b, c$  to obtain self-consistent equations. Thus,  $\tilde{Q}$  is a function of  $b, c$ , and on the other hand we can use  $\tilde{Q}$  to obtain an expression for  $b, c$  through the GPR formula in this two-dimensional subspace,

$$\begin{bmatrix} b \\ \sqrt{6}c \end{bmatrix} = \bar{\mathbf{t}} = \left[ \tilde{Q} + \frac{\sigma^2}{n} I_2 \right]^{-1} \mathbf{y} = \left[ \tilde{Q} + \frac{\sigma^2}{n} I_2 \right]^{-1} \begin{bmatrix} 1 \\ \sqrt{6}\epsilon \end{bmatrix} \quad (12)$$